VDF Classification: Investigation Progress

July 23, 2026

Abstract

This study reports preliminary results on clusterization of the Vlasiator 2D global simulation, with the idea of identification of the magnetospheric regions. The practical goal for now is to sanity-check the extraction, labelling, and PCA pipelines end to end before scaling to a full production run. Section 1 describes the expert-based region detection and the labelling strategies (116 VDFs across six substances). Section 2 reports blind PCA+KMeans clustering on the Hermite spectra of the VDFs, with lower-order physical moments (density, bulk velocity, anisotropic thermal velocity) added as explicit features. The best-scoring split ($k = 2$) recovers a cluster containing *all* `magnetosheath` and *all* `current_layer` samples (100% recall of both), at 81% purity within that cluster – these VDFs could be considered "disturbed" plasma. The complementary cluster is 100% pure calm-plasma labels.

Contents

1 Data processing

1.1 Smoke-test snapshot and VDF extraction

All results below are from `RUN_ID = smoke_test`: one local Vlasiator timestep (`bulk.0003408.vlsv`), a sparse, coarse 2D xz-plane run ($y \equiv 0$) chosen deliberately for fast pipeline iteration rather than physical realism – ∼2880 VDF-carrying cells over the whole domain, ∼2.35 R_E spacing. Extraction was restricted to a search box `SPATIAL_BOXRE = [-30, 30, -5, 5]` R_E (x, z), a band straddling the dayside magnetopause and the near tail. 116 VDF-carrying cells fell inside that box and were extracted with `VdfExtractor` at the snapshot's native velocity-mesh resolution ($4 \times 4 \times 4$ blocks/cell). Figure 1 shows their spatial distribution together with the labels assigned in Section 1.2.

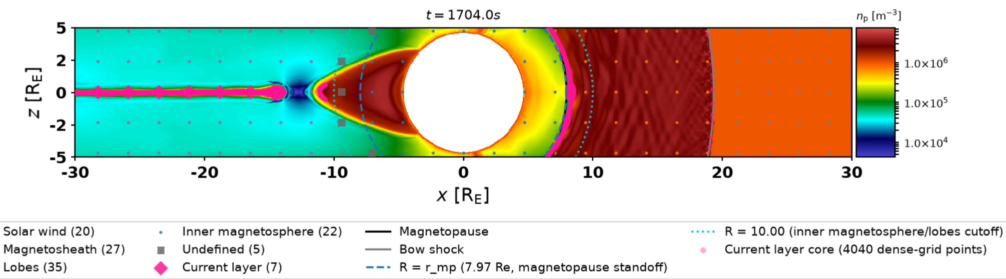

Figure 1: Extracted VDF cells, colored by ground-truth label, smoke_test run. Blue dashed circle: fitted Shue standoff r_{mp}. Cyan dotted circle: `lobe_r_min_re`. Pink dot cloud: the current-layer core's dense-grid extent; pink diamonds: the VDF cells actually labelled `current_layer`.

1.2 Labelling

1.2.1 Finding the magnetopause and bow shock

Both boundaries are located by a 1D density scan along the Sun-Earth line ($+x$, $y = z = 0$), from $x = 30\ R_E$ (undisturbed solar wind) inward to $x = 5\ R_E$ (near Earth), sampled at 300 points (`find_subsolar_point`). Moving inward, the density profile crosses two boundaries in sequence: first a steep *rise* (shocked, compressed solar wind – the bow shock), then, further in, a steep *drop* (from the dense sheath back down into the more tenuous magnetosphere – the magnetopause). Both crossings are found the same way: the magnetopause is the grid step with the steepest drop in $\log(\rho)$ over the whole scan; the bow shock is the steepest *rise* in $\log(\rho)$, restricted to the sunward side of the magnetopause crossing. On this run: magnetopause at $r_{mp} \approx 7.97\ R_E$ (density $1.13 \times 10^6 \rightarrow 1.85 \times 10^5$ m^{-3}), bow shock at $r_{bs} \approx 19.34\ R_E$ (density $1.10 \times 10^6 \rightarrow 2.86 \times 10^6$ m^{-3}, a ~3-4× compression – the theoretical maximum for a perpendicular MHD shock).

These two subsolar crossing distances anchor a subsolar Shue et al. (1998)-shaped surface, $r(\theta) = r_{mp} \left(\frac{2}{1+\cos\theta}\right)^{\alpha}$, with θ measured from the Sun-Earth line – one surface for the magnetopause (using r_{mp}) and one for the bow shock (using r_{bs}), both sharing the canonical flaring exponent $\alpha = 0.58$ (this run has no solar wind monitor to fit α from locally). This is a standard empirical parametrization of how both boundaries flare away from Earth off the Sun-Earth line, fitted here from a single subsolar density cut rather than a full boundary scan.

1.2.2 What each region is

With both flared surfaces in hand, every extracted VDF cell is classified by `classify_magnetosphere_regions` purely from its position (and local density) relative to them:

- **Solar wind**: outside the bow-shock surface – undisturbed upstream plasma that hasn't yet interacted with Earth's magnetosphere. A fast, narrow, close-to-Maxwellian beam.
- **Magnetosheath**: between the bow-shock and magnetopause surfaces – solar wind that has crossed the bow shock and been shocked, compressed, and heated. Denser and hotter than the solar wind, with a broader, more turbulent VDF (decelerated and deflected around the magnetopause).
- **Inner magnetosphere**: inside the magnetopause surface and close to Earth ($R < r_{mp}$, a simple sphere, not the flared shape) – dominated by Earth's dipole field, populated by trapped, quasi-thermal plasma.
- **Lobes**: inside the magnetopause surface but far from Earth ($R \geq$ `lobe_r_min_re`, 10 R_E here) – tenuous, open-field-line tail plasma, the coldest/most rarefied population in this taxonomy.
- **Current layer** (Section 2's "disturbed" population, together with magnetosheath): overrides whichever of the other regions a cell would otherwise get, wherever a cell coincides with a peak-$|J|$ core cell (the only active point substance this run, `core_fraction = 0.2`, dayside/tail search boxes independent, cells within 6 R_E of Earth excluded).
- **Undefined**: inside the magnetopause surface, in the geometric gap $r_{mp} \leq R <$ `lobe_r_min_re` $- r_{mp}$ is a *dayside-only* standoff distance, so applying it uniformly at all angles would mislabel near-Earth nightside plasma as lobes; `lobe_r_min_re` is deliberately set larger to avoid that, and the resulting band is left unassigned rather than guessed.

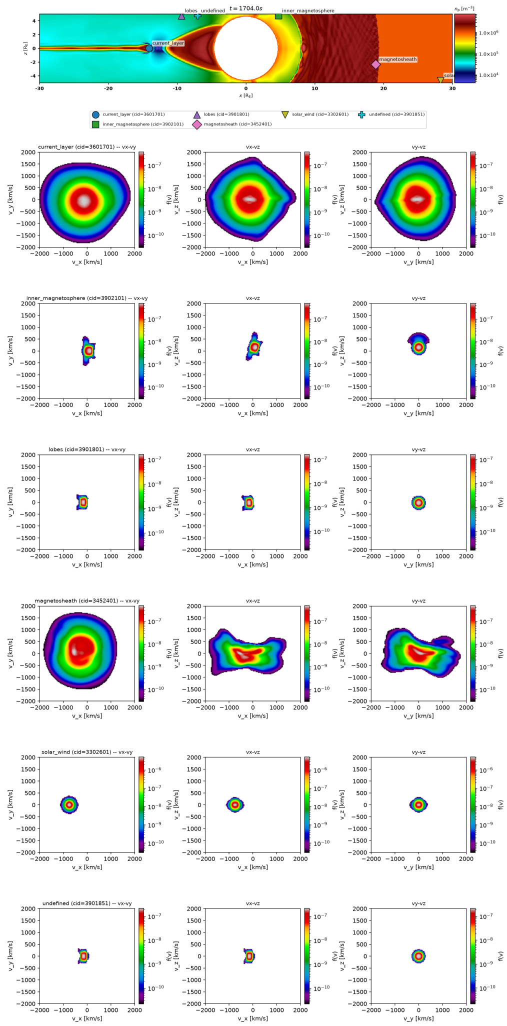

Figure 2: Labelling verification, smoke_test run: where each representative VDF sits (top) and its velocity-space cuts (v_x-v_y, v_x-v_z, v_y-v_z) below.

Table 1 gives the resulting counts.

One representative VDF cell per label (chosen uniformly at random) is shown in Figure 2: a header row with each representative's spatial position, followed by its three velocity-space cuts through its own peak, in the same label order – a sanity check that the label assigned to each representative matches the plasma population it physically sits in, and that its VDF shape is plausible for that population. The header and the cuts below it are generated together as one

Label	Count
lobes	35
magnetosheath	27
inner_magnetosphere	22
solar_wind	20
current_layer	7
undefined	5
Total	116

Table 1: Ground-truth label counts, smoke_test run.

figure (`plot_cluster_vdf_examples`, whose header row reuses `plot_cluster_vdf_positions`'s own drawing code) specifically so the two never need to be cross-referenced by hand.

1.3 Rotation and Hermite-spectrum processing

1.3.1 Rotation into the local magnetic-field frame

The same physical distribution can look very different in the simulation frame depending on where in the domain it sits, simply because the local magnetic field points in a different direction – an orientation confound that a shape-sensitive representation should not have to learn. Every extracted VDF is therefore resampled into its own field-aligned orthonormal frame

$$\hat{e}_{\parallel} = \hat{B}, \qquad \hat{e}_{\perp} = \frac{\mathbf{u} - (\mathbf{u}\cdot\hat{B})\,\hat{B}}{\left|\mathbf{u} - (\mathbf{u}\cdot\hat{B})\,\hat{B}\right|}, \qquad \hat{e}_{\perp 2} = \hat{e}_{\parallel} \times \hat{e}_{\perp}, \tag{1}$$

where $\mathbf{B}$ and $\mathbf{u}$ are the cell's local magnetic field and bulk velocity. The bulk velocity is Gram–Schmidt orthogonalized against $\hat{B}$ (rather than used raw) so the resulting matrix is an exact orthonormal rotation even in the presence of strong field-aligned flow; the degenerate case $\mathbf{u} \parallel \mathbf{B}$, where no perpendicular direction exists, falls back to the unrotated VDF and is counted rather than silently mis-rotated. The distribution is resampled onto the rotated grid by trilinear interpolation. Interpolation slightly smooths the distribution, so the recovered density is monitored as a rotation-correctness diagnostic: it changes by well under 1% (e.g. -0.36% for the representative in Figure 3), whereas a large shift would signal a broken rotation.

1.3.2 Log-space Hermite transform

In the rotated frame, each VDF is projected onto a separable basis of Hermite functions – the natural spectral basis for near-Maxwellian plasmas, whose zeroth mode *is* a drifting Maxwellian and whose higher modes encode progressively finer velocity-space structure. Along each axis the basis functions are

$$\psi_m(\xi) = \left(2^m m!\,\sqrt{\pi}\,v_{th}\right)^{-1/2} H_m(\xi)\,e^{-\xi^2/2}, \qquad \xi = \frac{v-u}{v_{th}}, \tag{2}$$

with H_m the physicists' Hermite polynomials; the drift velocity u and (isotropic) thermal speed v_{th}, which set the basis center and width, are genuine moments computed from the raw linear VDF. The spectrum is the tensor-product projection

$$H(m_{\parallel}, m_{\perp}, m_{\perp 2}) = \int g(\mathbf{v})\,\psi_{m_{\parallel}}(\xi_{\parallel})\,\psi_{m_{\perp}}(\xi_{\perp})\,\psi_{m_{\perp 2}}(\xi_{\perp 2})\,d^3v, \tag{3}$$

truncated at HERMITE_ORDER = 14 per axis, so each VDF becomes a $14 \times 14 \times 14$ coefficient array. Crucially, the projected quantity is not f itself but its *logarithm relative to the sparsity floor*,

$$g(\mathbf{v}) = \begin{cases} \log_{10}(f(\mathbf{v})/f_{\mathrm{sp}}), & f(\mathbf{v}) > f_{\mathrm{sp}}, \\ 0, & \text{otherwise,} \end{cases} \tag{4}$$

where f_{sp} is the simulation's own sparse-storage threshold. The logarithm keeps the wings' contribution comparable to the core's (a linear projection is dominated by the peak, hiding exactly the non-Maxwellian tail structure the classification targets), and subtracting the floor – rather than substituting $\log_{10} f_{\mathrm{sp}}$ as a constant background – keeps below-threshold cells at exactly zero, preserving the VDF's compact support: on this fixture ∼99.85% of the velocity grid is empty, and integrating a constant background over it would otherwise dominate every coefficient with a term unrelated to the VDF's actual shape.

Figure 3 shows the full processing chain for the magnetosheath representative: the raw VDF with the in-plane direction of **B** marked, the same distribution in the rotated $(v_\parallel, v_\perp)$ frame, and its Hermite spectrum reduced to two dimensions by summing in quadrature over the second perpendicular index, $H(m_\parallel, m_\perp) = [\sum_{m_{\perp 2}} H^2(m_\parallel, m_\perp, m_{\perp 2})]^{1/2}$ – the Hermite-space analogue of a gyrotropic reduction. The checkerboard pattern (alternating even/odd mode amplitudes decaying with order) is characteristic of the structured, shocked magnetosheath population; quiet near-Maxwellian populations concentrate almost entirely in the lowest few modes.

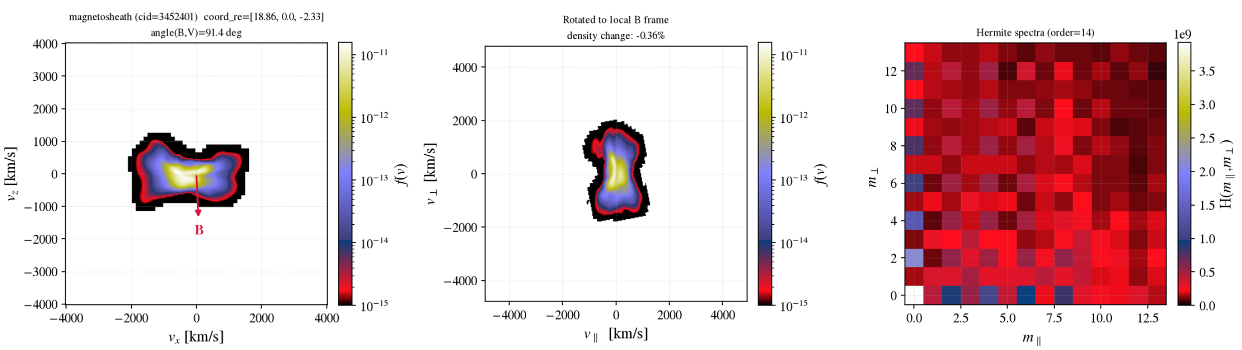

Figure 3: Rotation and Hermite processing chain for the magnetosheath representative (cid=3452401), smoke_test run. Left: raw VDF v_x–v_z slice, with the in-plane direction of **B** (red arrow) and the angle between **B** and the bulk flow annotated. Center: the same VDF resampled into its local field-aligned frame (Eq. 1); the density change from interpolation (−0.36%) is the rotation-correctness diagnostic. Right: its log-space Hermite spectrum (Eqs. 3–4), reduced to $(m_\parallel, m_\perp)$ by quadrature over the second perpendicular index.

Seven lower-order physical moments – density, 3-component bulk velocity, 3-component anisotropic thermal velocity – were computed in the same rotated frame and kept as separate features (Section 2.2), since the Hermite transform's own u/v_{th} normalization deliberately strips out absolute density/velocity/width.

2 PCA analysis

2.1 Feature representation

Results below use the "hermite" feature representation: each sample's $14 \times 14 \times 14$ log-space Hermite spectrum (Section 1) flattened directly into a 2744-dimensional feature vector – no slicing/downsampling, unlike the `"raw"`/`"rotated"` pixel-slice representations, since the Hermite basis already sees the full 3D VDF in a compact form.

2.2 Feature preparation

Three preparation steps were applied, in this order, before `StandardScaler`+PCA:

1. **Per-sample normalization** (`sample_normalization = "standard"`): each sample's flattened spectrum is mean-centered and divided by its own standard deviation. Physically structured populations (`current_layer`, `magnetosheath`) have raw spectrum magnitudes ∼6–9× larger than quieter populations at the *same shape* – `StandardScaler` alone (per-feature column scaling) leaves each sample's own overall scale intact, so without this step that scale difference dominates variance ahead of any shape difference.

2. **Moment features concatenated** (`include_moment_features = True`): the 7 moment columns from Section 1 are appended *after* the per-sample normalization above, and are *not* themselves per-sample-normalized – they are exactly the absolute-scale information the Hermite transform's u/v_{th} normalization discards, added back explicitly.

3. **Post-`StandardScaler` moment weighting** (`moment_feature_weight = 10.0`): after `StandardScaler` gives every column unit variance, the 7 moment columns are multiplied by 10×. Outnumbered ∼392:1 by Hermite columns (2744 vs. 7), their aggregate contribution to total variance would otherwise be diluted by column count alone, independent of how informative they are.

PCA was fit with `n_components = 40`; KMeans was fit for $k \in \{2, \ldots, 14\}$ and k selected by silhouette score.

2.3 Clustering results

The silhouette score peaked at $k = 2$ (0.314); scores for $k > 2$ were lower and comparable to each other (0.17–0.28, peaking again mildly at $k = 12$, 0.284) rather than showing a second clear elbow. Table 2 gives the $k = 2$ cluster composition against the ground-truth labels of Table 1.

Cluster	Size	lobes	magnetosheath	inner_mag.	solar_wind	current_layer	undefined
1 (disturbed)	42	1	27	6	0	7	1
0 (calm)	74	34	0	16	20	0	4

Table 2: $k = 2$ cluster composition, smoke_test run, hermite features + moments. Cluster 1 contains 100% of `magnetosheath` (27/27) and 100% of `current_layer` (7/7) – full recall of both "disturbed" substances – at 81% purity (34/42) within the cluster. Cluster 0 is 100% pure calm-plasma labels (`lobes/ solar_wind/inner_magnetosphere/undefined`), zero contamination from either disturbed substance.

Figures: `pca_scatter.png` (first two PCs, colored by both `cluster_ml` and `cluster_phys`), `silhouette_by_k.png`, `cluster_hermite_spectra.png` and `phys_cluster_hermite_spectra.png` (flattened Hermite spectra of representative samples, per PCA cluster and per physical label respectively), all under `data/plots/pca_snapshot/smoke_test/`.

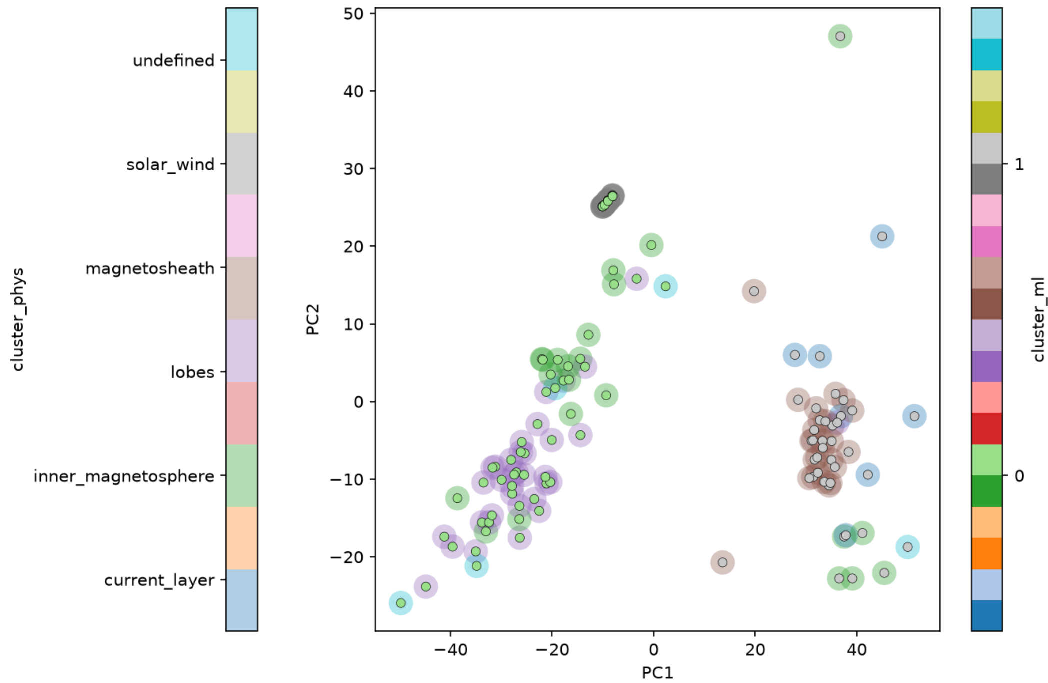

Figure 4: PCA scatter (first two principal components), smoke_test run, hermite features + moments, $k = 2$.

2.4 Discussion

With this feature set, blind clustering reliably separates “disturbed” plasma (`current_layer`, `magnetosheath` – structured, non-Maxwellian VDFs) from “calm” plasma (`lobes`/`solar_wind`/`inner_magnetosphe` `undefined` – near-Maxwellian) with *perfect recall* of both disturbed substances at $k = 2$: every `magnetosheath` and every `current_layer` sample lands in cluster 1, and cluster 0 has zero contamination from either. What it does *not* yet do is resolve substructure *within* either tier – silhouette scores for $k > 2$ don't show a clean second elbow, so KMeans on this feature set doesn't naturally separate `magnetosheath` from `current_layer`, or `lobes`/`solar_wind`/ `inner_magnetosphere` from each other. The 19% contamination in cluster 1 (6 `inner_magnetosphere` + 1 `lobes` + 1 `undefined`) is also worth a closer look at those specific cells' spectra – open question for the next iteration, not yet investigated here.

Appendix: run configuration

`src/data_proc/pipeline_config.py`, as run for this report (`RUN_ID = "smoke_test"`):

```
SPATIAL_BOXRE = [-30, 30, -5, 5]
POINTS_CONFIG["active_point_substances"] = ["current_layer"]
POINTS_CONFIG["current_layer_selection"]["core_fraction"] = 0.2
POINTS_CONFIG["current_layer_selection"]["min_r_re"] = 6.0
MAGNETOPAUSE_CONFIG["lobe_r_min_re"] = 10.0
BUILD_ROTATED_DATASET = False
BUILD_HERMITE_DATASET = True
HERMITE_ORDER = 14
```

```python
PCA_CONFIG = {
    "n_components": 40,
    "k_range": list(range(2, 15)),
    "feature_representation": "hermite",
    "sample_normalization": "standard",
    "include_moment_features": True,
    "moment_feature_weight": 10.0,
}
```